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EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS | Las Vegas

Time Travelling Data

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Level: Introductory

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Time-Travelling Data

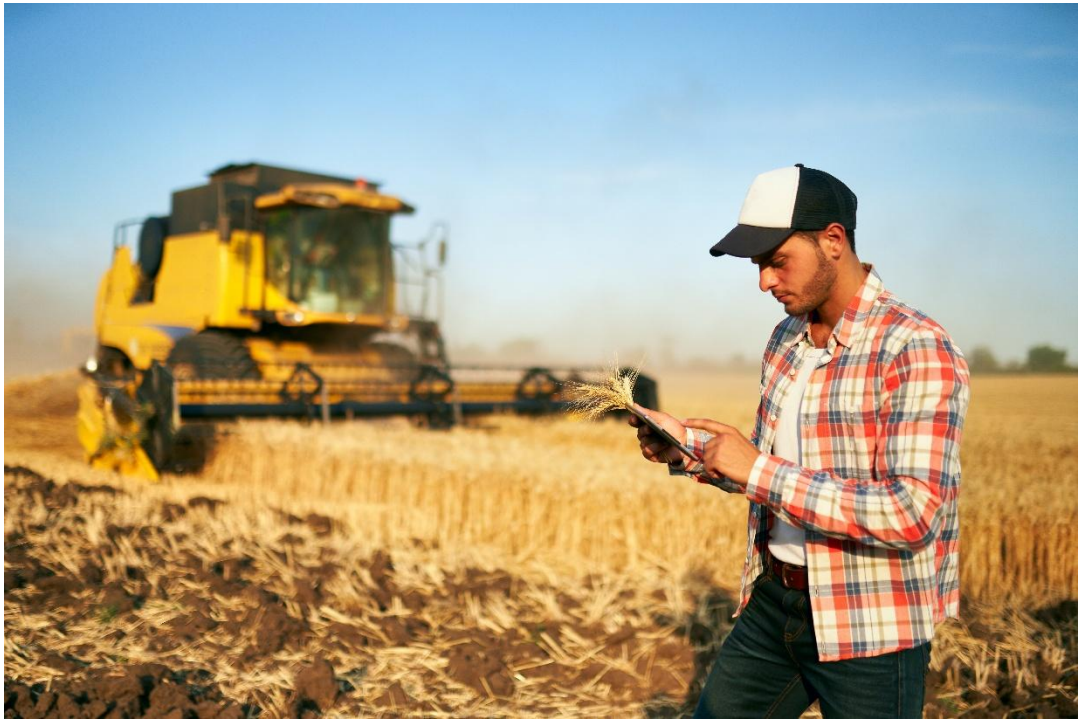
WHAT ARE SYSTEM-VERSIONED TEMPORAL TABLES?

Track Changes

What are system-versioned temporal tables?

System-Versioned

What are system-versioned temporal tables?

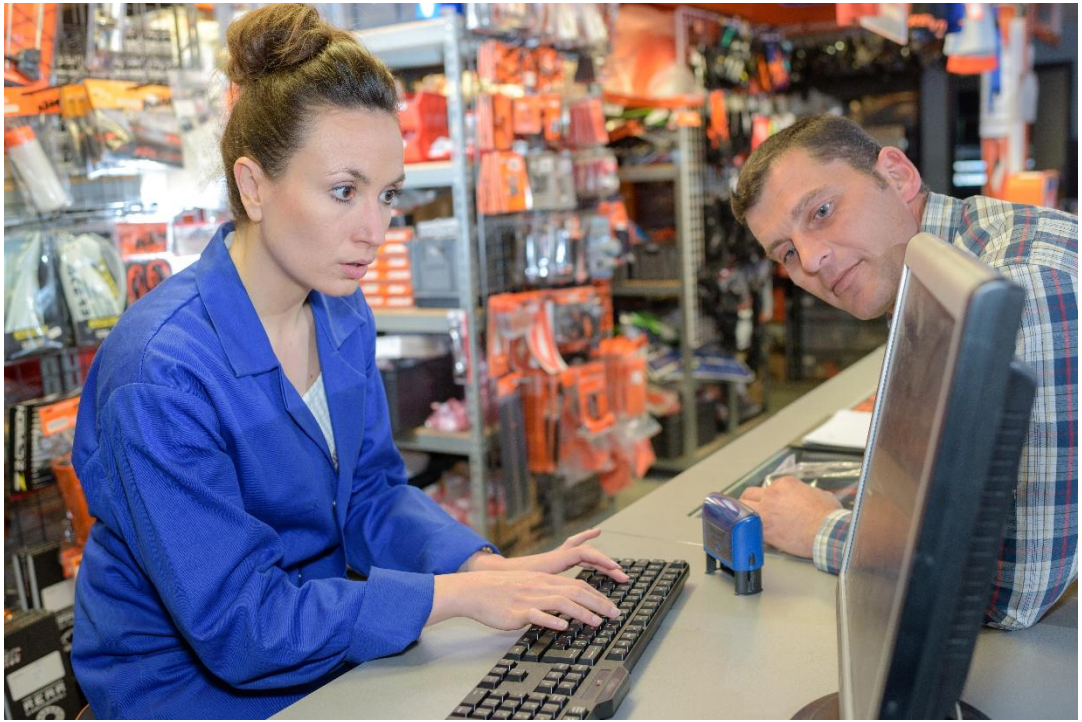


Rule of Two

What are system-versioned temporal tables?

Availability

What are system-versioned temporal tables?



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WHY USE TEMPORAL TABLES?

Why use temporal tables?

Immutable History

Built-In Time-Travelling Query Support

Better DML Performance

Minimal Maintenance Costs

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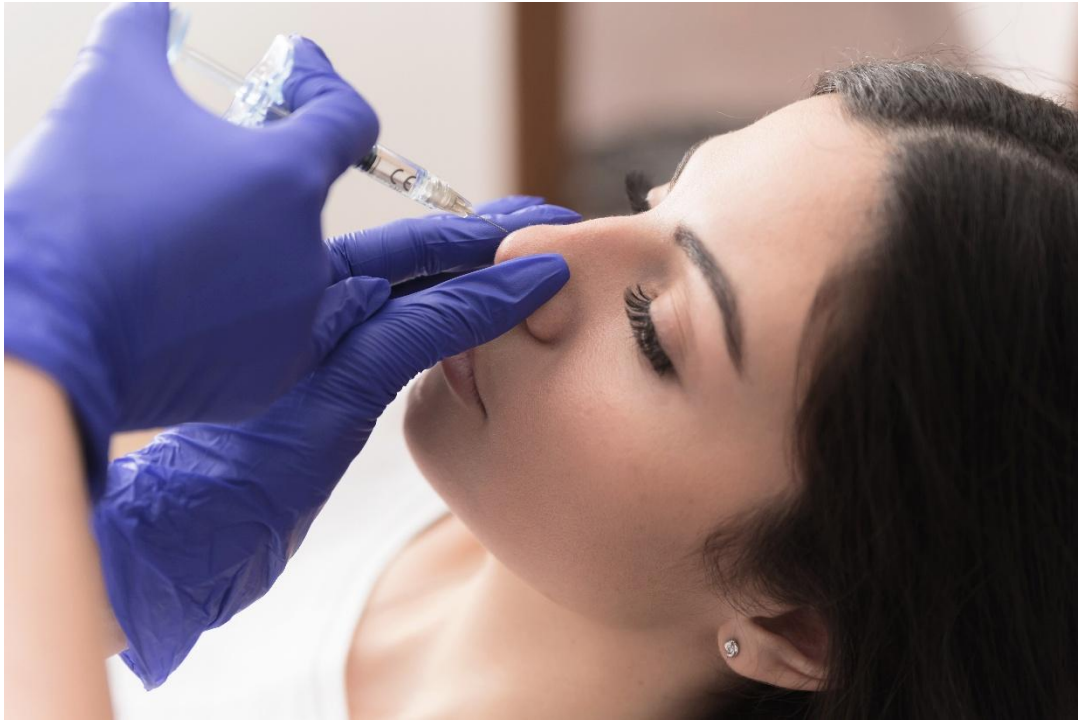
Auditing

Common Use Cases

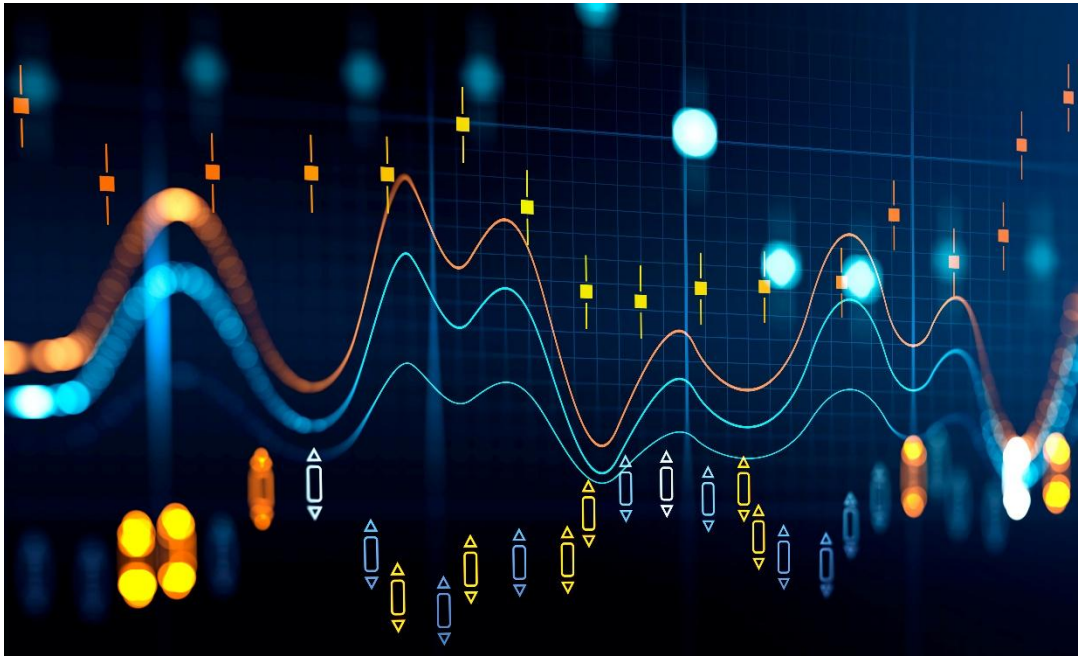
Reconstructing State

Common Use Cases



Calculating Trends

Common Use Cases



Slowly Changing Dimensions

Common Use Cases

Accidental Recovery

Common Use Cases

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HOW DO TEMPORAL TABLES WORK?

Ho do temporal tables work?

Database Table
PrimaryKeyId
Name
ForeignKeyId

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Ho do temporal tables work?

Temporal Table (Current)
PrimaryKeyId
Name
ForeignKeyId
ValidFrom
ValidTo

History Table
PrimaryKeyId
Name
ForeignKeyId
ValidFrom
ValidTo

Ho do temporal tables work?

Temporal Table (Current)	
PrimaryKeyId	123
Name	TaleLearnCode
ForeignKeyId	456
ValidFrom	2022-08-06 18:00:00
ValidTo	9999-12-31 23:59:59

History Table
PrimaryKeyId
Name
ForeignKeyId
ValidFrom
ValidTo

Insert

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Ho do temporal tables work?

Temporal Table (Current)	
PrimaryKeyId	123
Name	TaleLearnCode
ForeignKeyId	456
ValidFrom	2022-08-06 18:10:00
ValidTo	9999-12-31 23:59:59

History Table	
PrimaryKeyId	123
Name	TaleLearnCode
ForeignKeyId	456
ValidFrom	2022-08-06 18:00:00
ValidTo	2022-08-06 18:10:00

Update

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Ho do temporal tables work?

Temporal Table (Current)	
PrimaryKeyId	123
Name	[illegible]nChad
ForeignKeyId	
ValidFrom	2[illegible]6 18:10:00
ValidTo	9999-[illegible]2-31 23:59:59

History Table	
PrimaryKeyId	123
Name	BricksWithChad
ForeignKeyId	456
ValidFrom	2022-08-06 18:10:00
ValidTo	9999-12-31 23:59:59

Delete

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Ho do temporal tables work?

Temporal Table (Current)	
PrimaryKeyId	123
Name	BricksWithChad
ForeignKeyId	456
ValidFrom	2022-08-06 18:10:00
ValidTo	9999-12-31 23:59:59

History Table	
PrimaryKeyId	123
Name	BricksWithChad
ForeignKeyId	456
ValidFrom	2022-08-06 18:10:00
ValidTo	9999-12-31 23:59:59

Regular queries (current data)

Include historical version

Query

FOR SYSTEM_TIME
ALL, AS OF, BETWEEN..AND, FROM..TO, CONTAINED IN

Ho do temporal tables work?

```sql
SELECT * FROM DemoTable
FOR SYSTEM_TIME
BETWEEN '2022-08-05 00:00:00' AND '2022-08-06 23:59:59'
WHERE Id = 123
ORDER BY ValidFrom
```

ValidFrom 2022-08-06 18:10:00

ValidFrom 2022-08-06 18:10:00

PrimaryKeyId	Name	ForiegnKeyId	ValidFrom	ValidTo
123	BricskWithChad	456	2022-08-06 18:10:00	9999-12-31 23:59:59
123	TaleLearnCode	456	2022-08-06 18:00:00	2022-08-06 18:10:00

Query

FOR SYSTEM_TIME

ALL, AS OF, BETWEEN..AND, FROM..TO, CONTAINED IN

Ho do temporal tables work?

```
SELECT * FROM DemoTable
FOR SYSTEM_TIME
AS OF '2022-08-06 18:05:00'
WHERE Id = 123
```

ForeignKeyId	456
ValidFrom	2022-08-06 18:10:00

ForeignKeyId	456
ValidFrom	2022-08-06 18:10:00

PrimaryKeyId	Name	ForiegnKeyId	ValidFrom	ValidTo
123	TaleLearnCode	456	2022-08-06 18:00:00	2022-08-06 18:10:00

Regular queries (current data)

Include historical version

Query

FOR SYSTEM_TIME
ALL, AS OF, BETWEEN..AND, FROM..TO, CONTAINED IN

Time to
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TEMPORAL TABLES IN ENTITY FRAMEWORK

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Temporal T
Framewor

Support added in EF Core 6.0

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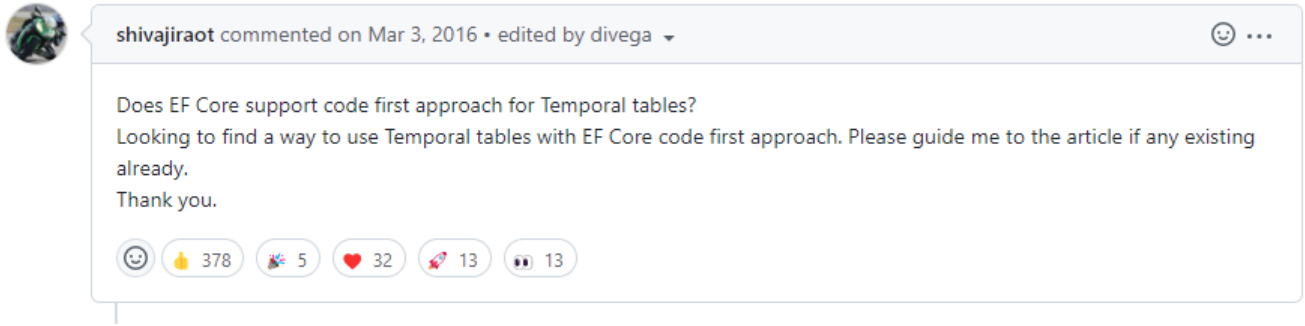
Temporal Tables in Entity Framework



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Temporal Tables in Entity Framework

Creation via EF Core Migrations

Transformation of Existing Tables

Querying Historical Data

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LINQ Extension Methods

TemporalAll

TemporalAsOf

TemporalFromTo

TemporalBetween

TemporalContainedIn

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